

C# Pattern Programs (20 Examples)

This document contains 20 different C# pattern programs. They are arranged from simple to advanced level. Each program includes explanation, code, and sample output.

1. Right Triangle Star Pattern

Prints a simple right-angled triangle of stars.

Code:

```
for(int i=1; i<=5; i++)
{
    for(int j=1; j<=i; j++)
        Console.Write("* ");
    Console.WriteLine();
}
```

Output:

```
*
* *
* * *
* * * *
* * * * *
```

2. Inverted Right Triangle

Stars in inverted right triangle form.

Code:

```
for(int i=5; i>=1; i--)
{
    for(int j=1; j<=i; j++)
        Console.Write("* ");
    Console.WriteLine();
}
```

Output:

```
* * * * *
* * * *
* * *
* *
*
```

3. Square Pattern

Stars arranged in square format.

Code:

```
for(int i=1; i<=5; i++)
{
    for(int j=1; j<=5; j++)
        Console.Write("* ");
    Console.WriteLine();
}
```

Output:

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

4. Hollow Square

Stars in a hollow square border.

Code:

```
for(int i=1; i<=5; i++)
{
    for(int j=1; j<=5; j++)
    {
        if(i==1 || i==5 || j==1 || j==5)
            Console.Write("* ");
        else
            Console.Write(" ");
    }
    Console.WriteLine();
}
```

Output:

```
* * * * *
*   *
*   *
*   *
* * * * *
```

5. Pyramid Pattern

Stars in a pyramid format.

Code:

```
for(int i=1; i<=5; i++)
{
    for(int j=1; j<=5-i; j++) Console.Write(" ");
    for(int k=1; k<=2*i-1; k++) Console.Write("*");
    Console.WriteLine();
}
```

Output:

```
      *
     ***
    *****
   *******
  *********
 ***********
```

6. Inverted Pyramid

Stars in inverted pyramid format.

Code:

```
for(int i=5; i>=1; i--)
{
    for(int j=1; j<=5-i; j++) Console.Write(" ");
    for(int k=1; k<=2*i-1; k++) Console.Write("*");
    Console.WriteLine();
}
```

Output:

```
*****
*****
*****
***
*
```

7. Diamond Pattern

Stars in diamond shape.

Code:

```
int n=5;
for(int i=1;i<=n;i++){
    for(int j=1;j<=n-i;j++) Console.Write(" ");
    for(int k=1;k<=2*i-1;k++) Console.Write("*");
    Console.WriteLine();
}
for(int i=n-1;i>=1;i--){
    for(int j=1;j<=n-i;j++) Console.Write(" ");
    for(int k=1;k<=2*i-1;k++) Console.Write("*");
    Console.WriteLine();
}
```

Output:

```

*
***
*****
*****
*****
*****
*****
***
*
```

8. Number Triangle

Numbers in increasing triangle pattern.

Code:

```
for(int i=1;i<=5;i++){
    for(int j=1;j<=i;j++) Console.Write(j+" ");
    Console.WriteLine();
}
```

Output:

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

9. Floyd's Triangle

Consecutive numbers in triangle form.

Code:

```
int num=1;
for(int i=1;i<=5;i++){
    for(int j=1;j<=i;j++) Console.Write(num++ + " ");
    Console.WriteLine();}
```

Output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

10. Pascal's Triangle

Pascal's triangle using binomial coefficients.

Code:

```
int n=5;
for(int i=0;i<n;i++){
    int val=1;
    for(int j=0;j<n-i;j++) Console.Write(" ");
    for(int k=0;k<=i;k++){
        Console.Write(val+" ");
        val=val*(i-k)/(k+1);
    }
    Console.WriteLine();}
```

Output:

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1